Prabin Kumar Rath

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EDUCATION

Ph.D. Computer Science — **Aug 2025 - May 2030**
Arizona State University, Tempe, AZ — 4.00 GPA

M.S. Artificial Intelligence and Robotics — **May 2024**
Arizona State University, Tempe, AZ — 4.00 GPA

B.S. Computer Science — **May 2020**
National Institute of Technology, Rourkela, India — 8.91 GPA

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C/C++, C#, TypeScript, Java, Shell
Cloud Technologies: Azure, AWS, Kubernetes, Docker **Simulators:** IsaacLab, PyBullet, CoppeliaSim, MuJoCo, Maniskill
Tools and Frameworks: PyTorch, TensorFlow, ROS/ROS2, OpenCV, Open3D, Git, Linux, Claude Code

PUBLICATIONS

[1] "Self-Supervised Keyframe Discovery for Horizon-Invariant Behavior Cloning", In ***NeurIPS 2026*** ↗
[2] "StageCraft: Execution Aware Mitigation of Distractor and Obstruction Failures in VLA Models", In ***IROS 2026*** ↗
[3] "Factorizing Diffusion Policies for Observation Modality Prioritization", In ***ICRA 2026*** ↗
[4] "XMoP: Whole-Body Control Policy for Zero-shot Cross-Embodiment Neural Motion Planning", In ***ICRA 2025*** ↗

RESEARCH EXPERIENCE

Vision-Language-Action (VLA) Models for Franka FR3 Robot | Systems Engineering — **Fall 2025 - Present**

- Extended the LeRobot backend to support the Franka FR3 robot platform through ROS2 APIs.
- Created a real-time teleoperation system with multiple control interfaces, including Meta Quest 3, GELLO whole-body leader-follower setup, and SpaceMouse.
- Developed robust and fault-tolerant robot teleoperation software for HRI user study experiments.
- Trained, deployed, and benchmarked multiple VLA and behavior cloning models, including $\pi_{0.5}$, SmolVLA, XVLA, Groot N1, DP, and ACT on the Franka FR3 robot, achieving 90%+ success rates on long-horizon manipulation tasks.

Policy Memory for Horizon-Invariant Behavior Cloning | NeurIPS 2026 — **Fall 2025 - Spring 2026**

- Proposed a novel self-supervised algorithm for memory-intensive behavior cloning that discovers sparse, decision-relevant keyframes in non-Markovian expert demonstrations without explicit keyframe annotations.
- Demonstrated behavior cloning with 100% SR at 1M-step horizons, outperforming SOTA baselines by 13.9% absolute SR on MIKASA-Robo and retaining 80% SR at 20× longer horizons on a real robot. 🌐 Project

Post-training Deployment Robustness under Visual Distribution Shifts | ICRA, IROS 2026 — **Fall 2024 - Spring 2026**

- Investigated prompting strategies for strict instruction following, spatial environment understanding, and low-hallucination failure reasoning in vision-language models (VLMs).
- Developed a training-free VLA improvement method leveraging VLM reasoning to anticipate and prevent failures, boosting success rates by 23% for $\pi_{0.5}$ and 54% for SmolVLA under visual disruptions and obstructions. 🌐 Project
- Improved Diffusion Policy robustness to visual distractors, camera occlusions, and appearance changes through sensor modality prioritization, outperforming baselines by 15% in simulation and 40% on real-world tasks. 🌐 Project

Cross-Embodiment Generalization in Neural Motion Planning | ICRA 2025 — **Fall 2023 - Spring 2024**

- Designed a novel behavior cloning policy architecture for neural motion planning, enabling zero-shot configuration-space generalization across 7 commercial robots with a 70% average planning success rate.
- Introduced a pointcloud-based semantic segmentation method for detecting collision points on robot surfaces, achieving cross-embodiment sim-to-real generalization with 76.6% IoU and 90.3% recall across 7 robots.
- Demonstrated sim-to-real motion planning from pointclouds on Franka FR3 and Rethink Sawyer robots, achieving a 72% average planning success rate in cluttered environments. 🌐 Project

INDUSTRY EXPERIENCE

Experian, Costa Mesa, CA: **MLOps Engineer** — **June 2024 - May 2025**

- Developed scalable workflows for end-to-end ML model lifecycle management and built inference orchestration platforms meeting millisecond-latency response SLAs while serving 300+ enterprise clients. 🌐 Product

Wells Fargo, Hyderabad, India: **Software Engineer** — **Aug. 2020 - July 2022**

- Architected and deployed robust, multithreaded REST API modules in Python, enabling efficient integration with Azure Cognitive Services. Optimized average response time by 75%, while delivering 4x throughput.

SERVICE

Paper Reviewer: CoRL 26 | ICRA 26, 25, 24 | RA-L 26, 23 | IROS 26, 23 | RO-MAN 23
Session Chair: Autonomous Vehicles @ ICRA 2025